Introduction to Slope-Intercept Form

Name _____ Date _____ Period _____

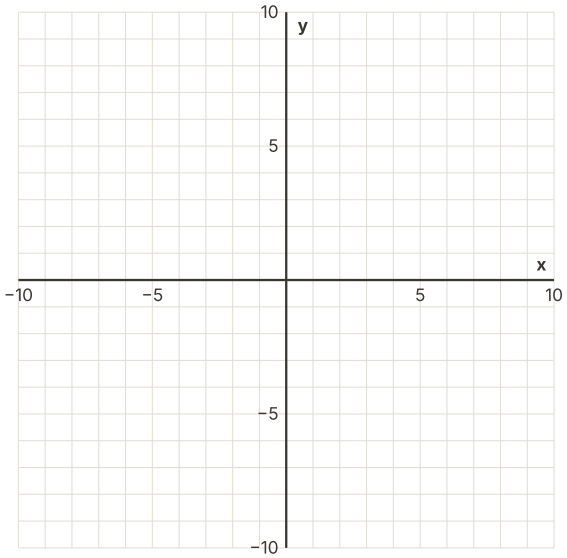
For each equation: fill in the table, plot the points on the graph, and draw the line through them.

Words to know

Slope-intercept form: y = mx + b.
Slope (m): how steep the line is. It is the number in front of x.
y-intercept (b): where the line crosses the y-axis, at (0, b).

1 y = x

Input (x)	y = x
−3	
−2	
−1	
0	
1	
2	
3	



✏ Practice: y = x

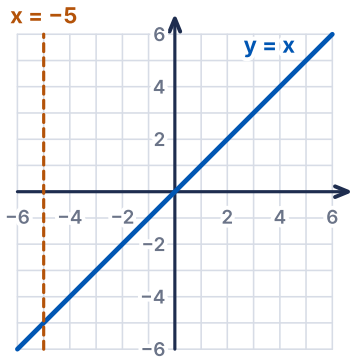
1. ★ Try it For y = x, what is y when x = 7?

x	y = x
0	0
1	1
2	2
7	?

Answer: _____

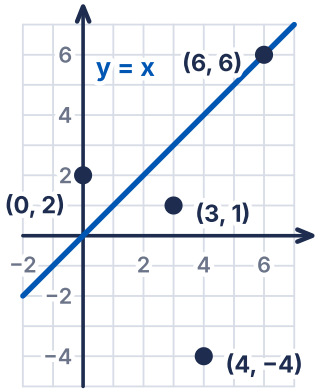
2. ★★ On level For y = x, what is y when x = −5?

Answer: _____



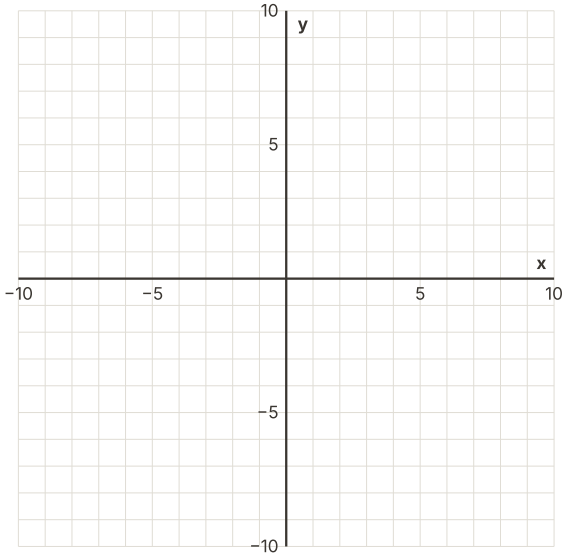
3. ★★★ Challenge Which point is on the line y = x?

- ○ (4, −4)
- ○ (6, 6)
- ○ (0, 2)
- ○ (3, 1)



2 y = −x

Input (x)	y = −x
−3	
−2	
−1	
0	
1	
2	
3	



✏ Practice: y = −x

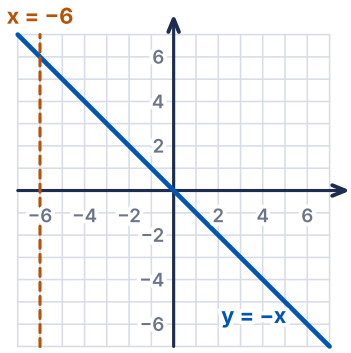
1. ★ Try it For y = −x, what is y when x = 4?

x	y = −x
0	0
1	−1
2	−2
4	?

Answer: _____

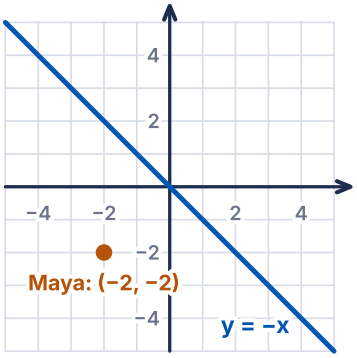
2. ★★ On level For y = −x, what is y when x = −6?

Answer: _____



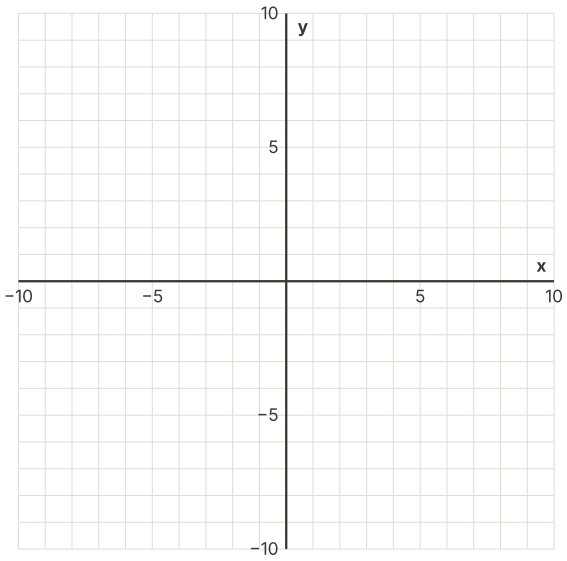
3. ★★★ Challenge Maya says the point (−2, −2) is on the line y = −x. Is she right?

- ○ Yes, she is right
- ○ No. When x = −2, y = 2
- ○ No. When x = −2, y = 0



3 y = x + 3

Input (x)	y = x + 3
−3	
−2	
−1	
0	
1	
2	
3	



✏ Practice: y = x + 3

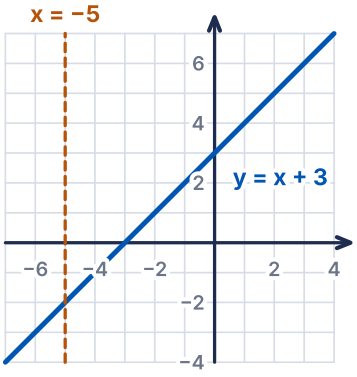
1. ★ Try it For y = x + 3, what is y when x = 2?

x	y = x + 3
0	3
1	4
2	?

Answer: _____

2. ★★ On level For y = x + 3, what is y when x = −5?

Answer: _____



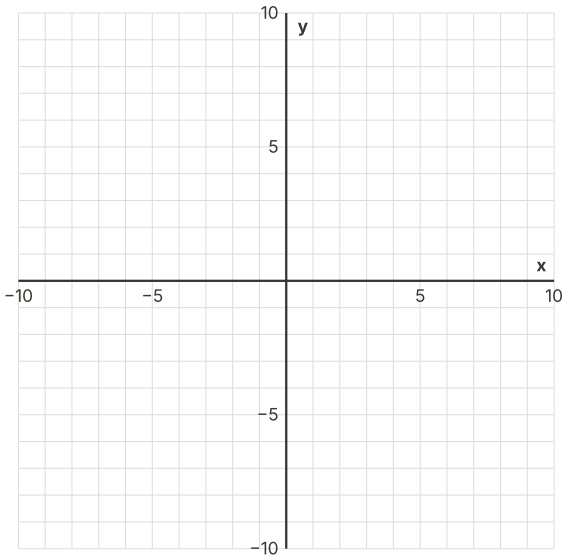
3. ★★★ Challenge On the line y = x + 3, what x-value gives y = 10?

x	y = x + 3
0	3
1	4
?	10

Answer: _____

4 y = x − 3

Input (x)	y = x − 3
−3	
−2	
−1	
0	
1	
2	
3	



✏ Practice: y = x − 3

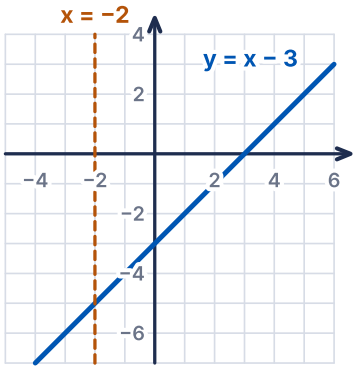
1. ★ Try it For y = x − 3, what is y when x = 5?

x	y = x − 3
0	−3
1	−2
5	?

Answer: _____

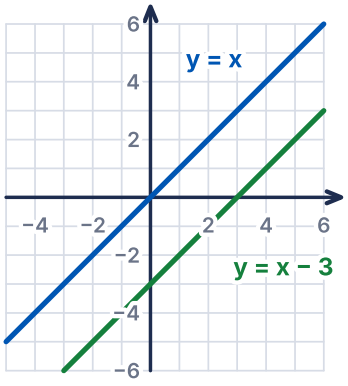
2. ★★ On level For y = x − 3, what is y when x = −2?

Answer: _____



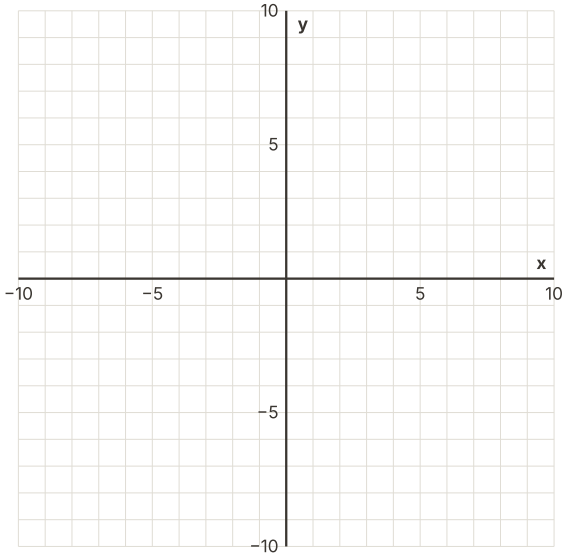
3. ★★★ Challenge Compare the graphs of y = x and y = x − 3. How is y = x − 3 different?

- ○ It is moved up 3
- ○ It is moved down 3
- ○ It is steeper
- ○ It goes down instead of up



5 $y = \frac{1}{2}x$

Input (x)	$y = \frac{1}{2}x$
−4	
−2	
0	
2	
4	
6	
8	



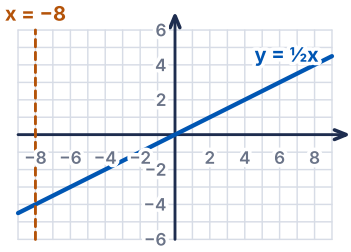
✏ Practice: y = ½x

1. ★ Try it For y = ½x, what is y when x = 10?

x	y = ½x
0	0
2	1
4	2
10	?

Answer: _____

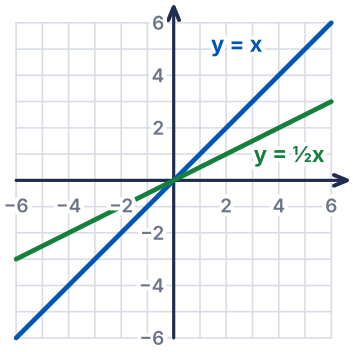
2. ★★ On level For y = ½x, what is y when x = −8?



Answer: _____

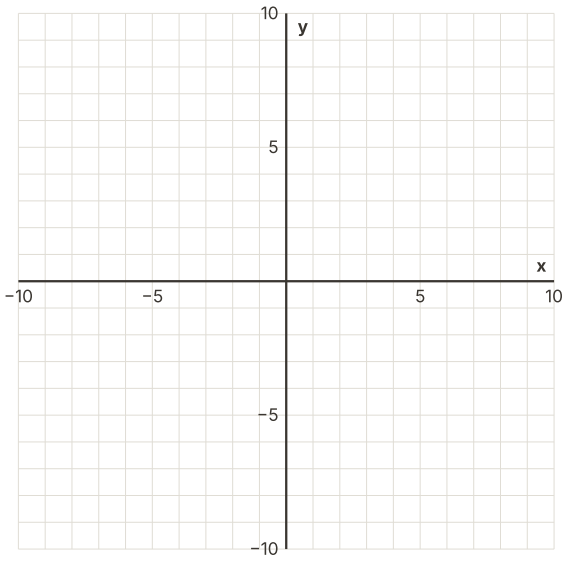
3. ★★★ Challenge Which line is steeper: y = x or y = ½x?

- ○ y = x
- ○ y = ½x
- ○ They are equally steep



6 $y = \frac{1}{2}x + 4$

Input (x)	$y = \frac{1}{2}x + 4$
−4	
−2	
0	
2	
4	
6	
8	



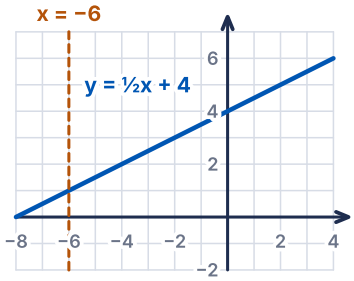
✏ Practice: y = ½x + 4

1. ★ Try it For y = ½x + 4, what is y when x = 2?

x	$y = \frac{1}{2}x + 4$
−2	3
0	4
2	?

Answer: _____

2. ★★ On level For y = ½x + 4, what is y when x = −6?



Answer: _____

3. ★★★ Challenge A plant's height after x weeks is y = ½x + 4 (in cm). How tall is it after 10 weeks? Write just the number.

Weeks (x)	Height in cm (y)
0	4
2	5
4	6
10	?

Answer: _____

Reflection Questions

A. Compare #1 and #2. What effect does the negative sign have on the graph?

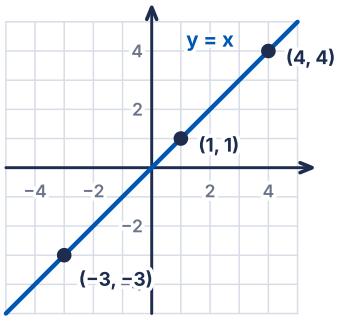
B. Compare #1, #3, and #4. What effect does adding or subtracting a number have?

C. Compare #1 and #5. What effect does changing the number in front of x have?

✏ Comprehensive Practice All sections

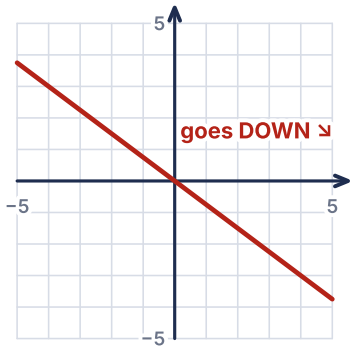
1. ★ Try it On the line y = x, the x-value and the y-value are always...

- ○ equal
- ○ opposites
- ○ 3 apart



2. ★ Try it Which line goes DOWN from left to right?

- ○ y = x
- ○ y = −x
- ○ y = x + 3
- ○ y = ½x



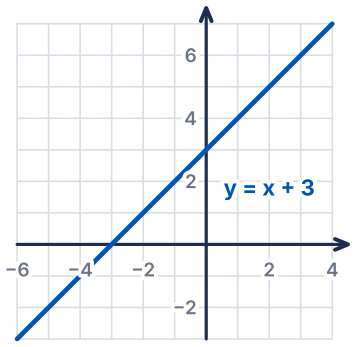
3. ★★★ Challenge Jordan says that on y = −x, when x = −3, y = −3. What is the correct y?

x	y = −x
−1	1
−2	2
−3	?

Answer: _____

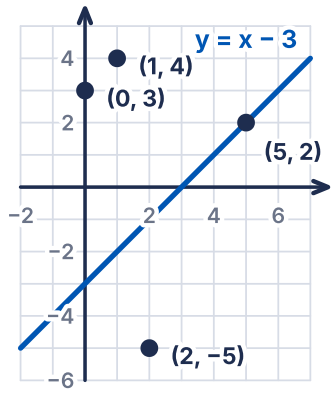
4. ★ Try it Where does the line y = x + 3 cross the y-axis? Write the y-value.

Answer: _____



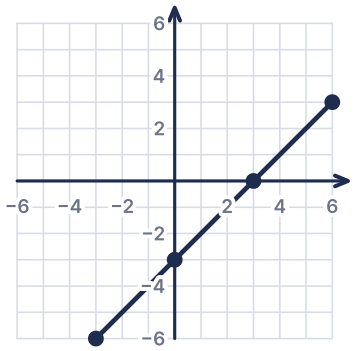
5. ★★ On level Which point is on the line y = x − 3?

- ○ (1, 4)
- ○ (5, 2)
- ○ (0, 3)
- ○ (2, −5)



6. ★★★ Challenge Which equation matches this graph?

- ○ y = x + 3
- ○ y = x − 3
- ○ y = −x − 3
- ○ y = ½x − 3



7. ★★ On level For y = ½x, what is y when x = 7? Write a decimal or a fraction.

x	y = ½x
2	1
4	2
6	3
7	?

Answer: _____

8. ★★★ Challenge Which equation matches this table?

- ○ y = x + 4
- ○ y = ½x + 4
- ○ y = ½x
- ○ y = 2x + 4

x	y
0	4
2	5
4	6
6	7

9. ★★ On level For y = −x + 2, what is y when x = 5?

Answer: _____

